

AI WeatherWise

Deliver intelligent weather forecasts and insights.

Project Documentation

Brainstorming to Documentation: 7 Phases

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Phase 1: Brainstorming & Ideation

Problem Statement

Raw weather data (temperature, rainfall, wind, pressure) is hard for most people to turn into decisions. People want to know whether to carry an umbrella, travel, water crops or schedule an event, not read numbers. AI WeatherWise turns forecast data into clear, personalised, actionable insight.

Vision

An AI-powered assistant that delivers accurate forecasts and plain-language insights tailored to the user's location and needs.

Ideas Considered

- Hyperlocal current conditions and 7-day forecast for any city
- Natural-language summaries ("Expect heavy rain after 4 PM, leave early")
- Activity-based advice: commute, travel, farming, outdoor events, fitness
- Severe-weather alerts (rain, heat, cyclone, storm) with safety tips
- Conversational Q&A ("Should I go for a run tomorrow?")
- Trend insights: comparisons with previous days and seasonal averages

Target Users

- Daily commuters and students
- Travellers and event planners
- Farmers and outdoor workers
- General public seeking simple forecasts

Unique Value

Combines live forecast data with AI that explains what the weather means and what to do about it.

Phase 2: Requirement Analysis

Functional Requirements

ID	Requirement	Priority
FR1	Search weather by city or use current location	High
FR2	Show current conditions and 5-7 day forecast	High
FR3	Generate AI insights and recommendations from forecast data	High
FR4	Display severe-weather alerts	Medium
FR5	Conversational question-answering about weather	Medium
FR6	Save favourite locations and units (°C/°F)	Low

Non-Functional Requirements

- Performance: results in under 3 seconds
- Reliability: graceful fallback if the weather API is unavailable
- Usability: clean, responsive interface for mobile and desktop
- Security: API keys stored server-side, never exposed in the client
- Accuracy: insights must be grounded in retrieved forecast data

Technical Requirements

- Weather data API (e.g. OpenWeatherMap or similar)
- AI/LLM service for insight generation
- Frontend web application; lightweight backend for API calls
- Version control and deployment hosting

Assumptions and Constraints

- API free-tier rate limits apply
- Internet connectivity is required
- Forecast accuracy depends on the data provider

Phase 3: Project Design

System Architecture

Three layers: (1) a web frontend for input and display, (2) a backend service that fetches weather data and calls the AI model, and (3) external services: the weather API and the AI model.

Component	Responsibility
Frontend UI	Location search, forecast cards, insights panel, chat box
Backend service	Validates requests, calls weather API, builds prompt, calls AI, returns result
Weather API	Supplies current, hourly and daily forecast data
AI model	Converts data into summaries, recommendations and answers
Cache	Stores recent results to reduce API calls and latency

Data Flow

- User enters a city or shares location
- Backend requests forecast data for that location
- Data is structured and sent to the AI model with clear instructions
- AI returns a plain-language insight; the UI displays forecast plus insight

UI Design

- Home: search bar, current weather card, AI insight banner
- Forecast view: hourly strip and daily cards
- Alerts: highlighted warnings with recommended actions
- Assistant: chat panel for follow-up questions

Phase 4: Project Planning

Timeline

Milestone	Tasks	Duration
M1 Setup	Repository, environment, API keys, project structure	Week 1
M2 Core	Location search, weather fetch, forecast display	Week 2
M3 AI	Prompt design, insight generation, alerts	Week 3
M4 Polish	Chat assistant, favourites, responsive UI	Week 4
M5 Test	Testing, bug fixes, performance checks	Week 5
M6 Release	Deployment and final documentation	Week 6

Risks and Mitigation

Risk	Mitigation
API downtime or rate limits	Caching and fallback message
AI gives inaccurate advice	Ground prompts in fetched data; show source data alongside insight
Scope creep	Prioritise High requirements first
Slow responses	Cache results, limit prompt size

Tools

Git/GitHub for version control, a task board for tracking, and a shared document for decisions.

Phase 5: Project Development

Approach

Development is iterative: build the smallest working slice (search to forecast), then layer AI insights, alerts and chat.

Modules

- **Weather module:** API client, response parsing, unit conversion
- **Insight module:** prompt template, AI call, output formatting
- **Alert module:** threshold rules for heavy rain, high heat, strong wind
- **UI module:** forecast cards, insight banner, chat interface
- **Utilities:** caching, error handling, configuration

Insight Prompt Design

The AI receives structured forecast data plus instructions: summarise in two or three sentences, give one practical recommendation, mention risks, and never state facts not present in the data.

Development Practices

- Small, frequent commits with clear messages
- Secrets in environment variables
- Consistent code style and inline comments for key logic

Phase 6: Project Testing

Test Strategy

- Unit tests: parsing, unit conversion, alert thresholds
- Integration tests: backend to weather API to AI model
- UI tests: responsiveness and navigation across devices
- AI output review: check insights match the underlying data

Sample Test Cases

ID	Scenario	Expected Result
TC1	Search valid city	Current weather and forecast displayed
TC2	Search invalid city	Friendly error message, no crash
TC3	Weather API unavailable	Fallback message shown; cached data used if present
TC4	Heavy rain in forecast	Alert shown with recommended action
TC5	Switch °C to °F	All values convert correctly
TC6	Ask "Do I need an umbrella tomorrow?"	Answer consistent with forecast data
TC7	Mobile screen width	Layout adapts without horizontal scrolling

Exit Criteria

All High-priority requirements pass; no critical bugs open; average response time under 3 seconds.

Phase 7: Project Documentation

Deliverables

- This project document covering all seven phases
- README with setup and run instructions
- API and configuration notes (environment variables, keys)
- User guide with screenshots
- Test report summarising results

Setup Summary

- Clone the repository and install dependencies
- Add weather API and AI service keys to environment configuration
- Start the backend and frontend
- Open the app and search a city

Future Enhancements

- Push notifications for severe weather
- Multi-language support
- Air quality and UV index insights
- Personalised profiles (farmer, commuter, traveller)
- Voice assistant

Conclusion

AI WeatherWise bridges the gap between raw forecast data and everyday decisions by pairing reliable weather sources with clear AI-generated guidance.